

Submission B solution to Problem 3

Overview / Strategy

The problem is to determine for which probabilities p the displayed inequality holds uniformly over all finite convex weight vectors. The main positive argument has two reductions. First, Hoeffding's extremal theorem for convex combinations of i.i.d. random variables reduces the weighted problem to the special case of equal weights. In the Bernoulli case, equal weights give binomial averages. Second, a classical binomial mean-tail estimate of Jogdeo–Samuels type says that for $0 \leq p \leq 1/3$, every binomial average is at least its mean with probability at least p . These two facts are stated precisely in Section 1 and used in Proposition 5.

The remaining cases are elementary. At $p = 1/2$, the Bernoulli subset is uniformly distributed over all subsets, and pairing every subset with its complement gives probability at least $1/2$. At $p = 1$, the assertion is immediate. To show that there are no other probabilities, equal weights on three variables give counterexamples for every $p \in (1/3, 1/2)$, and equal weights on two variables give counterexamples for every $p \in (1/2, 1)$.

Definition 1. A Bernoulli(p) random variable is a random variable taking value 1 with probability p and value 0 with probability $1 - p$. For $p \in [0, 1]$, let $\mathcal{P}(p)$ denote the following assertion: for every integer $m \geq 1$, every choice of nonnegative real numbers

$$w_1, \dots, w_m \geq 0, \quad \sum_{i=1}^m w_i = 1,$$

and every family $v_1, \dots, v_m$ of i.i.d. Bernoulli(p) random variables, one has

$$\Pr \left[\sum_{i=1}^m w_i v_i \geq p \right] \geq p.$$

Theorem 2 (Headline theorem: the original problem). *The assertion $\mathcal{P}(p)$ holds exactly for*

$$p \in [0, \tfrac{1}{3}] \cup \{\tfrac{1}{2}, 1\}.$$

Equivalently, the complete set of probabilities p for which

$$\Pr \left[\sum_{i=1}^m w_i v_i \geq p \right] \geq p$$

holds for every finite convex weight vector $(w_1, \dots, w_m)$ and every i.i.d. Bernoulli(p) family $v_1, \dots, v_m$ is

$$\boxed{[0, \frac{1}{3}] \cup \{\frac{1}{2}, 1\}}.$$

1 Classical one-dimensional inputs

The proof uses two standard one-dimensional results. They are quoted in the precise forms needed below. The first is Hoeffding's reduction theorem for convex combinations of i.i.d. random variables; the second is the binomial mean-tail estimate used to evaluate the equal-weight case.

Lemma 3 (Hoeffding's reduction for convex combinations). *Let $X_1, X_2, \dots$ be i.i.d. real-valued random variables. For $t \in \mathbb{R}$, define*

$$L(t) := \inf_{N \geq 1} \Pr \left[\frac{X_1 + \dots + X_N}{N} \geq t \right].$$

Then for every integer $m \geq 1$ and every choice of weights

$$w_1, \dots, w_m \geq 0, \quad \sum_{i=1}^m w_i = 1,$$

one has

$$\Pr \left[\sum_{i=1}^m w_i X_i \geq t \right] \geq L(t).$$

Proof. This is the one-dimensional half-line form of Hoeffding's extremal theorem for convex combinations of i.i.d. random variables; see Hoeffding [1]. To make explicit that the closed-tail convention used here causes no ambiguity, we record the standard passage from the open-tail formulation.

Hoeffding's theorem gives, for every real u ,

$$\Pr \left[\sum_{i=1}^m w_i X_i > u \right] \geq \inf_{N \geq 1} \Pr \left[\frac{X_1 + \dots + X_N}{N} > u \right].$$

Fix $t \in \mathbb{R}$. For every $\varepsilon > 0$, apply the preceding inequality with $u = t - \varepsilon$. For each N ,

$$\left\{ \frac{X_1 + \dots + X_N}{N} \geq t \right\} \subseteq \left\{ \frac{X_1 + \dots + X_N}{N} > t - \varepsilon \right\},$$

and therefore

$$\inf_{N \geq 1} \Pr \left[\frac{X_1 + \dots + X_N}{N} > t - \varepsilon \right] \geq \inf_{N \geq 1} \Pr \left[\frac{X_1 + \dots + X_N}{N} \geq t \right] = L(t).$$

Thus

$$\Pr \left[\sum_{i=1}^m w_i X_i > t - \varepsilon \right] \geq L(t) \quad \text{for every } \varepsilon > 0.$$

As $\varepsilon \downarrow 0$, the events

$$\left\{ \sum_{i=1}^m w_i X_i > t - \varepsilon \right\}$$

decrease to

$$\left\{ \sum_{i=1}^m w_i X_i \geq t \right\}.$$

By continuity of probability from above,

$$\Pr \left[\sum_{i=1}^m w_i X_i \geq t \right] \geq L(t).$$

This is the desired closed-tail statement. □

Lemma 4 (Binomial mean-tail inequality). *Let*

$$X_{N,p} \sim \text{Bin}(N, p).$$

Then the following hold.

1. *If $0 \leq p \leq \frac{1}{3}$, then for every $N \geq 1$,*

$$\Pr[X_{N,p} \geq Np] \geq p.$$

Equivalently, since $X_{N,p}$ is integer-valued,

$$\Pr[X_{N,p} \geq \lceil Np \rceil] \geq p.$$

2. *If $p = \frac{1}{2}$, then for every $N \geq 1$,*

$$\Pr[X_{N,1/2} \geq N/2] \geq \frac{1}{2}.$$

Moreover, in the first assertion equality occurs for $N = 1$ and $0 < p \leq 1/3$.

Proof. For $p = 0$, $X_{N,0} = 0$ almost surely, so

$$\Pr[X_{N,0} \geq 0] = 1 \geq 0.$$

For $0 < p \leq 1/3$, the first assertion is the classical binomial mean-tail estimate of Samuels and Jogdeo–Samuels; see [3, 2]. In the form needed here, that estimate states exactly that for every $N \geq 1$,

$$\Pr[X_{N,p} \geq \lceil Np \rceil] \geq p.$$

Because $X_{N,p}$ is integer-valued, the events

$$\{X_{N,p} \geq Np\} \quad \text{and} \quad \{X_{N,p} \geq \lceil Np \rceil\}$$

are identical. This proves the first assertion.

For $p = 1/2$, the binomial distribution is symmetric:

$$\Pr[X_{N,1/2} = j] = \Pr[X_{N,1/2} = N - j] \quad (0 \leq j \leq N).$$

If $N = 2r + 1$ is odd, then the values $0, \dots, r$ pair with the values $2r + 1, \dots, r + 1$, and exactly one member of each pair is at least $N/2 = r + \frac{1}{2}$. Hence

$$\Pr[X_{N,1/2} \geq N/2] = \frac{1}{2}.$$

If $N = 2r$ is even, then the values $0, \dots, r - 1$ pair with $2r, \dots, r + 1$, while r is the central value and is included in the event $\{X_{N,1/2} \geq r\}$. Therefore

$$\Pr[X_{N,1/2} \geq N/2] = \frac{1}{2}(1 - \Pr[X_{N,1/2} = r]) + \Pr[X_{N,1/2} = r] = \frac{1}{2} + \frac{1}{2} \Pr[X_{N,1/2} = r] \geq \frac{1}{2}.$$

This proves the second assertion.

Finally, if $N = 1$ and $0 < p \leq 1/3$, then

$$\{X_{1,p} \geq p\} = \{X_{1,p} = 1\},$$

so

$$\Pr[X_{1,p} \geq p] = p.$$

Thus equality is possible in the first assertion. □

2 The positive cases

Proposition 5 (The interval $0 \leq p \leq 1/3$). *The assertion $\mathcal{P}(p)$ holds for every*

$$p \in [0, \tfrac{1}{3}].$$

Proof. Fix $p \in [0, 1/3]$. Let $m \geq 1$, let

$$w_1, \dots, w_m \geq 0, \quad \sum_{i=1}^m w_i = 1,$$

and let $v_1, \dots, v_m$ be i.i.d. Bernoulli(p) random variables.

Extend $v_1, \dots, v_m$, if necessary, to an infinite i.i.d. Bernoulli(p) sequence $X_1, X_2, \dots$ with $X_i = v_i$ for $1 \leq i \leq m$. Applying Lemma 3 with $t = p$ gives

$$\Pr\left[\sum_{i=1}^m w_i v_i \geq p\right] = \Pr\left[\sum_{i=1}^m w_i X_i \geq p\right] \geq \inf_{N \geq 1} \Pr\left[\frac{X_1 + \dots + X_N}{N} \geq p\right].$$

For each N , the sum $X_1 + \cdots + X_N$ has binomial distribution $\text{Bin}(N, p)$. Hence, writing $X_{N,p} \sim \text{Bin}(N, p)$,

$$\Pr\left[\frac{X_1 + \cdots + X_N}{N} \geq p\right] = \Pr[X_{N,p} \geq Np].$$

Therefore

$$\Pr\left[\sum_{i=1}^m w_i v_i \geq p\right] \geq \inf_{N \geq 1} \Pr[X_{N,p} \geq Np].$$

By Lemma 4, since $0 \leq p \leq 1/3$,

$$\Pr[X_{N,p} \geq Np] \geq p \quad \text{for every } N \geq 1.$$

Taking the infimum over N gives

$$\inf_{N \geq 1} \Pr[X_{N,p} \geq Np] \geq p.$$

Consequently,

$$\Pr\left[\sum_{i=1}^m w_i v_i \geq p\right] \geq p.$$

Since m , the weights, and the Bernoulli(p) family were arbitrary, $\mathcal{P}(p)$ holds. $\square$

Proposition 6 (The point $p = 1/2$). *The assertion $\mathcal{P}(1/2)$ holds.*

Proof. Let $m \geq 1$, let

$$w_1, \dots, w_m \geq 0, \quad \sum_{i=1}^m w_i = 1,$$

and let $v_1, \dots, v_m$ be i.i.d. Bernoulli($1/2$) random variables.

Let

$$A = \{i \in \{1, \dots, m\} : v_i = 1\}.$$

Because the v_i are independent and each is 1 with probability $1/2$, the random set A is uniformly distributed over all 2^m subsets of $\{1, \dots, m\}$. For a subset $B \subseteq \{1, \dots, m\}$, write

$$w(B) = \sum_{i \in B} w_i.$$

Then

$$\sum_{i=1}^m w_i v_i = w(A),$$

so the desired event is

$$\{w(A) \geq 1/2\}.$$

Pair every subset $B \subseteq \{1, \dots, m\}$ with its complement B^c . These pairs partition the 2^m subsets into 2^{m-1} unordered pairs. Since

$$w(B) + w(B^c) = \sum_{i=1}^m w_i = 1,$$

at least one of B and B^c has weight at least $1/2$. Thus at least one subset in each complementary pair belongs to the event $\{w(A) \geq 1/2\}$.

Every subset has probability 2^{-m} . Since at least 2^{m-1} subsets satisfy $w(B) \geq 1/2$, we obtain

$$\Pr \left[\sum_{i=1}^m w_i v_i \geq \frac{1}{2} \right] = \Pr[w(A) \geq 1/2] \geq 2^{m-1} \cdot 2^{-m} = \frac{1}{2}.$$

Therefore $\mathcal{P}(1/2)$ holds. □

Proposition 7 (The point $p = 1$). *The assertion $\mathcal{P}(1)$ holds.*

Proof. Let $m \geq 1$, let

$$w_1, \dots, w_m \geq 0, \quad \sum_{i=1}^m w_i = 1,$$

and let $v_1, \dots, v_m$ be i.i.d. Bernoulli(1) random variables. A Bernoulli(1) random variable equals 1 almost surely, so

$$v_i = 1 \quad \text{almost surely for every } i.$$

Therefore

$$\sum_{i=1}^m w_i v_i = \sum_{i=1}^m w_i = 1$$

almost surely. Hence

$$\Pr \left[\sum_{i=1}^m w_i v_i \geq 1 \right] = 1 \geq 1.$$

Thus $\mathcal{P}(1)$ holds. □

3 Counterexamples outside the claimed set

Proposition 8 (Failure for $1/3 < p < 1/2$). *For every*

$$p \in \left(\frac{1}{3}, \frac{1}{2} \right),$$

the assertion $\mathcal{P}(p)$ fails.

Proof. Fix $p \in (1/3, 1/2)$. Take

$$m = 3, \quad w_1 = w_2 = w_3 = \frac{1}{3},$$

and let v_1, v_2, v_3 be i.i.d. Bernoulli(p) random variables. Put

$$N = v_1 + v_2 + v_3.$$

Then $N \sim \text{Bin}(3, p)$ and

$$\sum_{i=1}^3 w_i v_i = \frac{N}{3}.$$

Because $p > 1/3$, one success is not enough:

$$\frac{1}{3} < p.$$

Because $p < 1/2 < 2/3$, two successes are enough:

$$\frac{2}{3} \geq p.$$

Thus

$$\left\{ \sum_{i=1}^3 w_i v_i \geq p \right\} = \{N \geq 2\}.$$

Therefore

$$\Pr \left[\sum_{i=1}^3 w_i v_i \geq p \right] = \Pr[N \geq 2].$$

Since $N \sim \text{Bin}(3, p)$,

$$\Pr[N \geq 2] = \binom{3}{2} p^2(1-p) + p^3 = 3p^2(1-p) + p^3 = p^2(3-2p).$$

Now

$$p - p^2(3-2p) = p(1-3p+2p^2) = p(1-p)(1-2p).$$

For $p \in (1/3, 1/2)$, the factors p , $1-p$, and $1-2p$ are all positive. Hence

$$p - p^2(3-2p) > 0,$$

or equivalently,

$$p^2(3-2p) < p.$$

Thus this choice of weights gives

$$\Pr \left[\sum_{i=1}^3 w_i v_i \geq p \right] < p.$$

Therefore $\mathcal{P}(p)$ fails. □

Proposition 9 (Failure for $1/2 < p < 1$). *For every*

$$p \in \left(\frac{1}{2}, 1\right),$$

the assertion $\mathcal{P}(p)$ fails.

Proof. Fix $p \in (1/2, 1)$. Take

$$m = 2, \quad w_1 = w_2 = \frac{1}{2},$$

and let v_1, v_2 be i.i.d. Bernoulli(p) random variables. Put

$$N = v_1 + v_2.$$

Then

$$\sum_{i=1}^2 w_i v_i = \frac{N}{2}.$$

Because $p > 1/2$, one success is not enough:

$$\frac{1}{2} < p.$$

Since N can only take the values 0, 1, 2, the event

$$\left\{ \frac{N}{2} \geq p \right\}$$

occurs only when $N = 2$, that is, when both v_1 and v_2 equal 1. Hence

$$\Pr \left[\sum_{i=1}^2 w_i v_i \geq p \right] = \Pr[v_1 = 1, v_2 = 1].$$

By independence,

$$\Pr[v_1 = 1, v_2 = 1] = p^2.$$

Since $0 < p < 1$,

$$p^2 < p.$$

Thus this choice of weights gives

$$\Pr \left[\sum_{i=1}^2 w_i v_i \geq p \right] < p.$$

Therefore $\mathcal{P}(p)$ fails. □

4 Proof of the classification

Proof of Theorem 2. Proposition 5 proves that $\mathcal{P}(p)$ holds for every

$$p \in [0, \tfrac{1}{3}].$$

Proposition 6 proves that $\mathcal{P}(1/2)$ holds, and Proposition 7 proves that $\mathcal{P}(1)$ holds. Hence

$$\mathcal{P}(p) \text{ holds for every } p \in [0, \tfrac{1}{3}] \cup \{\tfrac{1}{2}, 1\}.$$

It remains to show that no other probability works. Since p is a probability, $p \in [0, 1]$. The complement of

$$[0, \tfrac{1}{3}] \cup \{\tfrac{1}{2}, 1\}$$

inside $[0, 1]$ is

$$\left(\tfrac{1}{3}, \tfrac{1}{2}\right) \cup \left(\tfrac{1}{2}, 1\right).$$

For every $p \in (1/3, 1/2)$, Proposition 8 gives an explicit choice of weights for which the desired inequality fails. For every $p \in (1/2, 1)$, Proposition 9 gives an explicit choice of weights for which the desired inequality fails.

Therefore $\mathcal{P}(p)$ holds exactly for

$$p \in [0, \tfrac{1}{3}] \cup \{\tfrac{1}{2}, 1\}.$$

This proves the theorem. □

References

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- [2] K. Jogdeo and S. M. Samuels, *Monotone convergence of binomial probabilities and a generalization of Ramanujan's equation*, Annals of Mathematical Statistics **39** (1968), 1191–1195.
- [3] S. M. Samuels, *On the number of successes in independent trials*, Annals of Mathematical Statistics **36** (1965), 1272–1278.
- [4] A. Fernandez, *Convex Combinations of Low Probability Bernoulli Variables*, Math.StackExchange discussion, <https://math.stackexchange.com/questions/265163/convex-combinations-of-low-probability-bernoulli-variables>.